Oxford College of Emory University

Chapter 14, Part 1

Money, Liquidity, and Banking

Guilherme Paiva Pinto

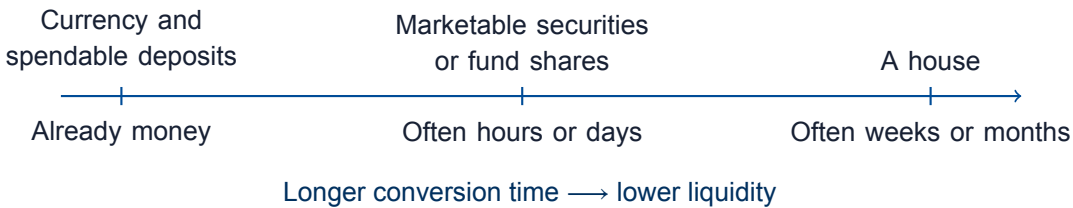
October 27, 2026

Roadmap

1. Money, liquidity, and the money supply
2. Banks, reserves, and T-accounts
3. Bank runs and safeguards
4. Moral hazard and shadow banking

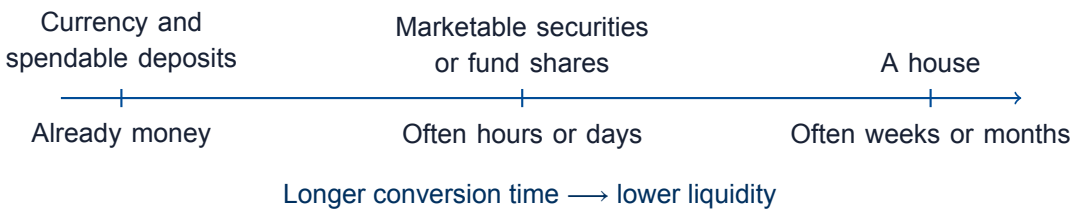
Money and Liquidity

- **Money**: assets that can readily be used to purchase goods and services.



Money and Liquidity

- **Money**: assets that can readily be used to purchase goods and services.
- **Liquidity**: how quickly an asset can become spendable money, with little cost or loss of value.



The Money Supply: M1 and M2

- **Money supply**: the total value of financial assets counted as money.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Money Supply: M1 and M2

- **Money supply**: the total value of financial assets counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.

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The Money Supply: M1 and M2

- **Money supply**: the total value of financial assets counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.
- **M2, the broader measure**: M1 plus small time deposits and retail money market mutual fund balances.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.

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  - Pay for groceries without a double coincidence of wants.
- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.
- **Store of value**: a way to carry purchasing power into the future.
  - Save today's earnings for next month's groceries. Inflation can erode their value.

Types of Money

A farmer brings produce to market. Compare three ways to pay.



Producing crops: Bruegel, *The Harvesters*, 1565



Trading: Van Ruysdael, *Market by the Seashore*, 1637

Commodity money

Pay with a portion of the crop accepted as money. It also has value as food.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

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Commodity-backed money
Pay with a warehouse receipt redeemable for a specified quantity of grain.

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Fiat money
Pay with officially accepted money that promises no commodity redemption.

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What Banks Do

- Banks **accept deposits** and provide payment services.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

What Banks Do

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- They keep part of their funding in **reserves** and lend a large portion or invest it in other assets.

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What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or invest it in other assets.
- Deposits can be withdrawn quickly; many loans and investments repay over years.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

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- Reserves let a bank meet withdrawals and settle payments to other banks.

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Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.
- Reserves let a bank meet withdrawals and settle payments to other banks.
- **Reserve ratio**: bank reserves divided by the bank's deposits.

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T-Accounts

A **T-account** lists **assets on the left** and **liabilities on the right** to summarize a bank or business.

Assets		Liabilities	
Loans	$1,200,000	Deposits	$1,000,000
Reserves	$100,000		

Assets: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' equity.

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

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Reserve ratio $= \frac{100{,}000}{1{,}000{,}000} = 10\%.$

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Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

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Reserves +$100; deposits +$100.

Money and Banking Checkpoint

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What changes for the bank and for public currency + deposits?

Reserves +$100; deposits +$100.

Public currency −$100, so currency + deposits is unchanged.

Bank Runs

- A **bank run** occurs when many depositors withdraw because they fear the bank may fail.

Source: Federal Reserve Board, SVB review, April 2023.

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Bank Runs

- A **bank run** occurs when many depositors withdraw because they fear the bank may fail.
- The bank holds only part of its funding as immediately available reserves; loans cannot all be collected at once.
- Forced asset sales can cause losses, increasing fear and triggering more withdrawals.

Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: March 2023

- **2022**: higher rates reduced long-term securities' values; technology-sector customers drew down deposits.

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- **March 8–9, 2023**: SVB announced securities losses and a capital plan; the next day, deposits fell by more than $40 billion.
- **March 10, 2023**: California closed the bank. Many large, uninsured and connected depositors made the run especially fast.

Source: Federal Reserve Board, SVB review, April 2023.

Bank Regulation

- **Deposit insurance**: FDIC created in 1933; coverage began in 1934. Eligible deposits: up to $250,000 per depositor, insured bank, and ownership category.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

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- **Interest on reserve balances (IORB)**: the Fed pays interest on eligible reserve balances and helps guide short-term rates.
- **Capital requirements**: a minimum cushion of owners' equity relative to exposures absorbs losses.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.

Source: Federal Reserve Board, regulatory capital.

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- Deposit insurance can weaken depositors' incentive to monitor a bank. Owners may gain from risky bets while others bear part of the losses.

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Moral Hazard

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- Deposit insurance can weaken depositors' incentive to monitor a bank. Owners may gain from risky bets while others bear part of the losses.
- **Capital requirements** put more of owners' funds at risk; supervision complements that incentive.

Source: Federal Reserve Board, regulatory capital.

Shadow Banking

- **Shadow banking**: credit intermediation outside ordinary deposit banking.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

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- Examples: **investment banks**, **money market funds**, hedge or credit funds, finance companies, and securitization vehicles.

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Shadow Banking

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- Examples: **investment banks**, **money market funds**, hedge or credit funds, finance companies, and securitization vehicles.
- Some insurance-group activities also belong to this chain. Short-term funding of illiquid assets can create run-like risks.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Shadow Banking in the Great Recession

Date	Institution	What happened
March 2008	Bear Stearns Investment bank	Funding run; acquired with Fed support.
September 15, 2008	Lehman Brothers Investment bank	Filed for bankruptcy.
September 16, 2008	AIG Insurance group	Collateral pressure; emergency Fed support.
September 16, 2008	Reserve Primary Fund Money market fund	Lehman losses; share value fell below $1.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

Source: Federal Reserve Board, AIG assistance, September 16, 2008.

Money and Bank Claims

- A deposit is money to its holder and a liability of the bank.

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- Reserves support payments; equity absorbs losses.
- Part 2: the Fed, loan creation, settlement, repayment, and deposit expansion.

Chapter 14, Part 2

Oxford College of Emory University

Money Creation and Central Banks

Guilherme Paiva Pinto

October 29, 2026

Roadmap

1. Money creation and the deposit multiplier
2. The monetary base and money supply
3. The Federal Reserve: structure and tools
4. The European Central Bank

Determining the Money Supply

Money supply = public currency + checkable deposits

Simplified measure: M = C + D

The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing			

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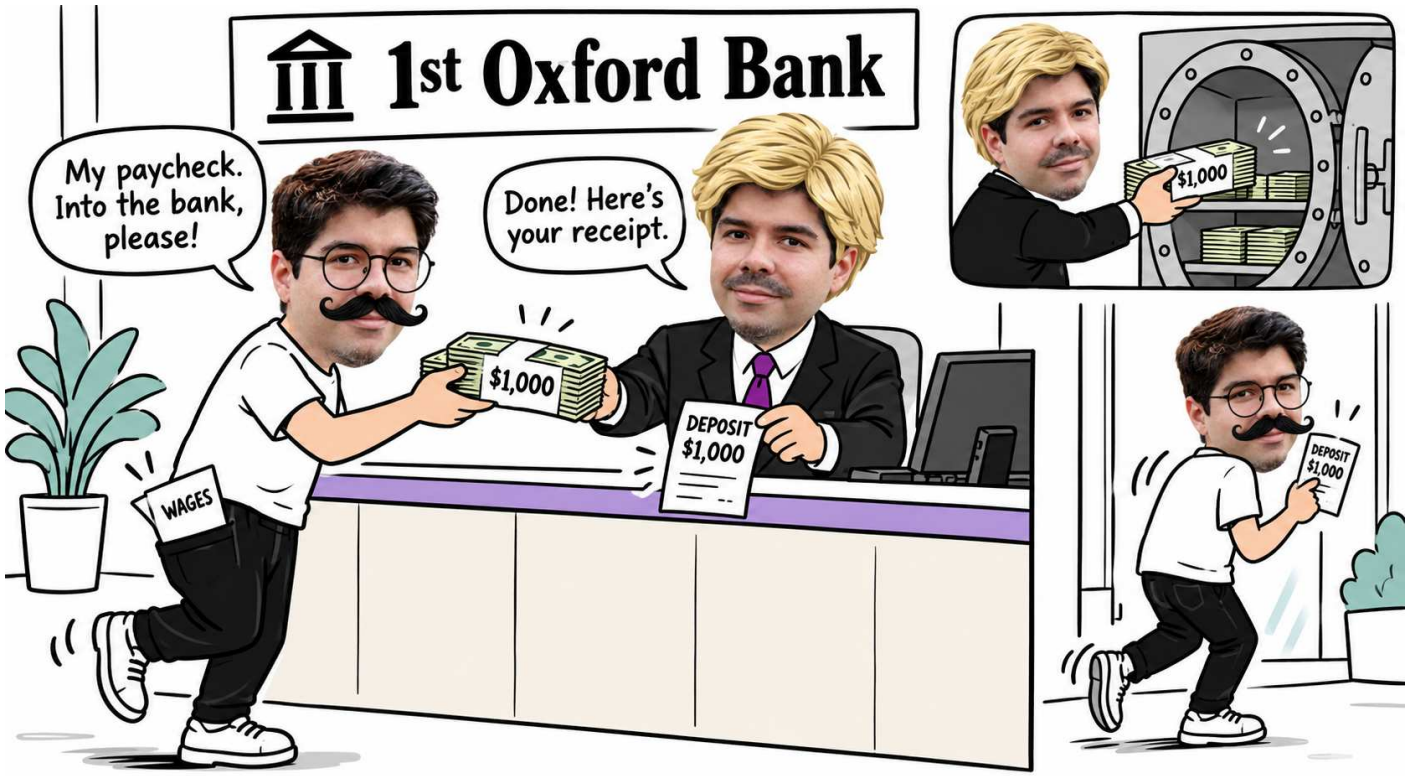
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The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing	$0	$1,000	$1,000

The deposit changes the form of money, not its total.

How Banks Create Money

Starting wages: $1,000 cash · Model reserve requirement: rr = 10%



Wages deposited: $1,000.

Money-supply accounting ($)

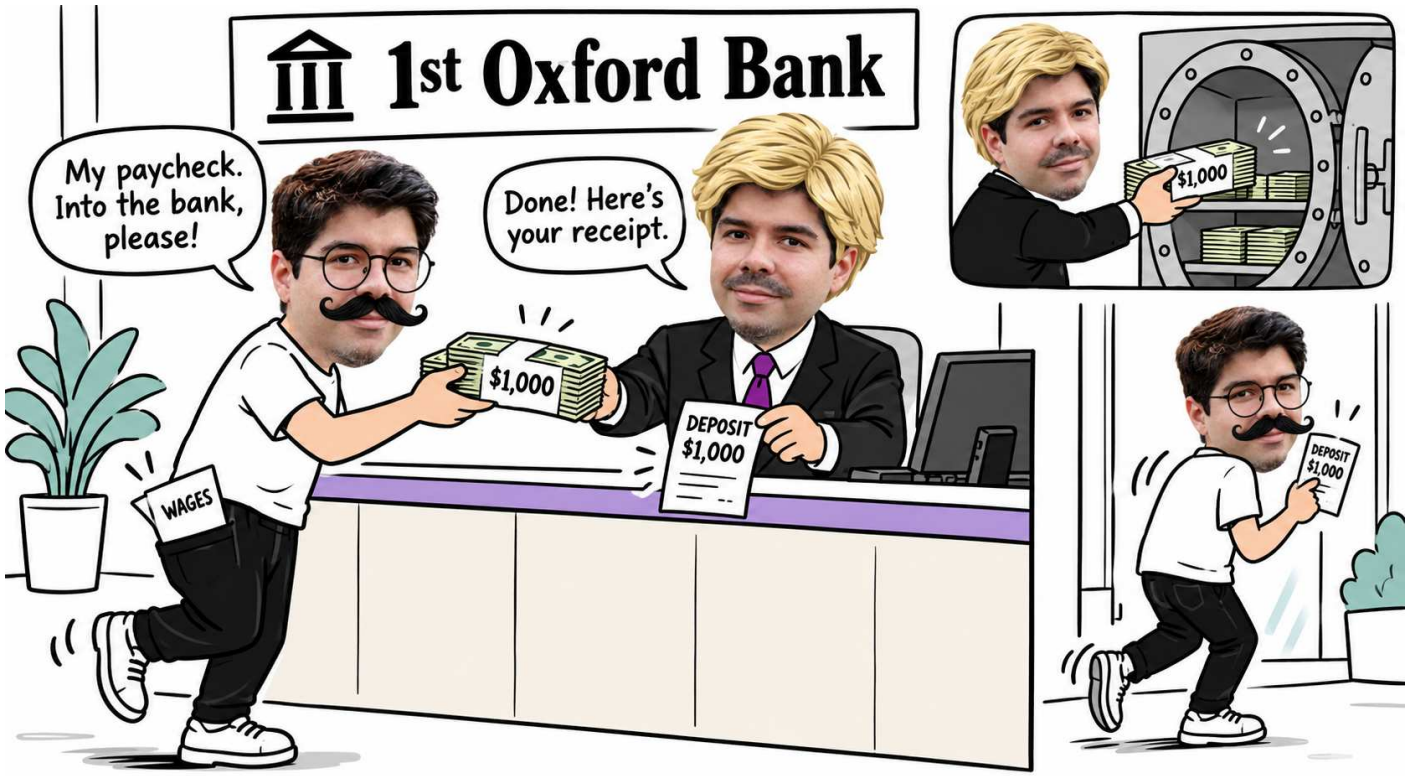
Step	C	D	M
Before	1,000	0	1,000
Deposit	0	1,000	1,000
Loan credit			
Cash / spend			
Redeposit			

D: deposits · C: public currency

M = C + D

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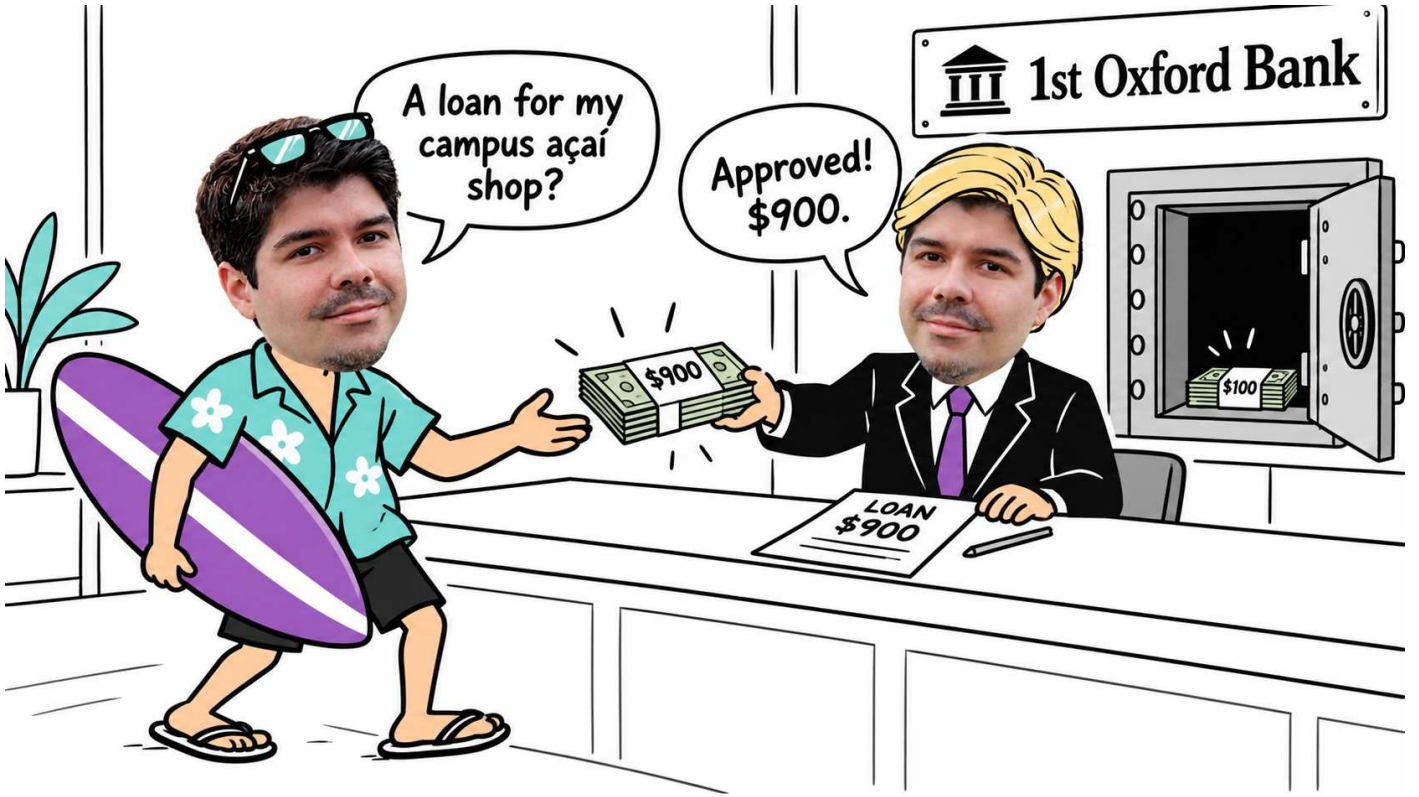
D: deposits · C: public currency

M = C + D

Required reserves = $100; excess reserves = $900.

How Banks Create Money

Starting wages: $1,000 cash · Model reserve requirement: rr = 10%



Loan credit creates $900; taking cash changes its form.

Money-supply accounting ($)

Step	C	D	M
Before	1,000	0	1,000
Deposit	0	1,000	1,000
Loan credit	0	1,900	1,900
Cash / spend	900	1,000	1,900
Redeposit			

D: deposits · C: public currency

M = C + D

How Banks Create Money

Starting wages: $1,000 cash · Model reserve requirement: rr = 10%



Spending passes the same $900 to the supplier.

Money-supply accounting ($)

Step	C	D	M
Before	1,000	0	1,000
Deposit	0	1,000	1,000
Loan credit	0	1,900	1,900
Cash / spend	900	1,000	1,900
Redeposit			

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Starting wages: $1,000 cash · Model reserve requirement: rr = 10%



Money-supply accounting ($)

Step	C	D	M
Before	1,000	0	1,000
Deposit	0	1,000	1,000
Loan credit	0	1,900	1,900
Cash / spend	900	1,000	1,900
Redeposit	0	1,900	1,900

D: deposits · C: public currency

M = C + D

The supplier deposits $900 at Bank B.

Lending adds money; payment and redeposit do not add it again.

Reserves and Repeated Deposit Creation

Excess reserves = reserves − required reserves.

rr = 10% · Each row follows lending and payment to the next bank.

Bank	Deposits	Reserves kept	New loan
A	$1,000		
B			
C			
…	…	…	…

Reserves and Repeated Deposit Creation

Excess reserves = reserves − required reserves.

At Bank A: $1,000 − (0.10 × $1,000) = **$900**.

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Bank	Deposits	Reserves kept	New loan
A	$1,000	$100	$900
B	$900	$90	$810
C			
…	…	…	…

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C	$810	$81	$729
…	…	…	…

The Simple Money Multiplier

Initial cash deposit: \$1,000 · rr = 0.10

$$D = 1{,}000 + 1{,}000(1 - rr) + 1{,}000(1 - rr)^2 + \ldots$$

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$10,000 total deposits includes the original $1,000.

$9,000 additional deposits are created through lending.

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Under these assumptions, \$1 of reserves supports \$10 of deposits.

Net increase in C + D from before the cash deposit: \$9,000.

Why Reality Is More Complicated

Full redeposit. People redeposit all loan proceeds.

No excess reserves. Banks lend the amount allowed by the model.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

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Full redeposit. People redeposit all loan proceeds.

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Borrowers qualify. Willing, creditworthy borrowers want the loans.

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Why Reality Is More Complicated

Full redeposit. People redeposit all loan proceeds.

No excess reserves. Banks lend the amount allowed by the model.

Borrowers qualify. Willing, creditworthy borrowers want the loans.

More public cash or excess reserves → less deposit expansion.

1/rr is a teaching model, not an automatic prediction.
The 10% rule is hypothetical; U.S. reserve requirements have been 0% since March 2020.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

Monetary Base and Money Supply

Monetary base

$$MB = C + R$$

Public currency + bank reserves

Money supply

$$M = C + D$$

Public currency + checkable deposits

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Reserves = vault cash + balances at the Fed.

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Bank reserves are not public deposits.

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$$\text{Money multiplier} = \frac{M}{MB}$$

Next: how Fed operations change reserves and the monetary base.

The Federal Reserve System



1. Boston
2. New York
3. Philadelphia
4. Cleveland
5. Richmond
6. Atlanta
7. Chicago
8. St. Louis
9. Minneapolis
10. Kansas City
11. Dallas
12. San Francisco

Reserve Bank locations. San Francisco also serves Alaska and Hawaii.

Board of Governors

7 seats · Washington, D.C.

Nominated by the President; confirmed by the Senate.

Federal Reserve, System Structure · Map: Natural Earth

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Board of Governors

7 seats · Washington, D.C.

Nominated by the President; confirmed by the Senate.

Oversees the System. Governors serve staggered 14-year terms.

Federal Reserve, System Structure · Map: Natural Earth

The Federal Open Market Committee

FOMC = the Fed's monetary-policy committee.

7	+	1	+	4
Board governor seats		New York Fed president		Rotating Reserve Bank presidents

Federal Reserve, FOMC · Federal Reserve, System Structure

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All Reserve Bank presidents participate; voting rights rotate.
The FOMC does not set every household or business loan rate.

Federal Reserve, FOMC · Federal Reserve, System Structure

What the Fed Does

Monetary policy · financial stability and supervision · payments

The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

New York Fed, Monetary Policy Implementation

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FFR: the federal funds rate—the interest rate in this market.

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Policy tool	What changes
Discount rate	Cost of borrowing from the Fed
Interest on reserve balances	Return on eligible balances at the Fed
Open-market operations	Securities and bank reserve balances

Next week: interest rates, spending, and aggregate demand.

New York Fed, Monetary Policy Implementation

The Discount Window and Reserve Interest

Discount window

The Fed lends to eligible banks against collateral.

Interest on reserve balances

IORB pays banks on eligible balances held at the Fed.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

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Cheaper Fed funding; borrowing becomes more attractive.

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A reservation rate:
a safe alternative return to lending.

Higher IORB raises that benchmark; lower IORB reduces it.

IORB influences lending rates; it is not a fixed minimum rate on every loan.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

Open-Market Purchases

The Fed buys a $1,000 Treasury security from a commercial bank.



Purchase: reserves enter the bank.

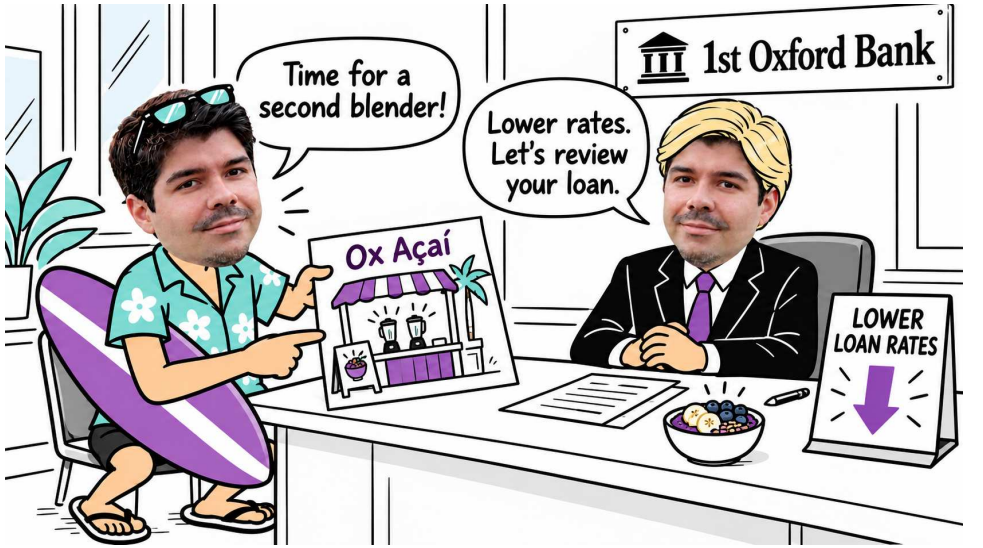
New York Fed, Monetary Policy Implementation

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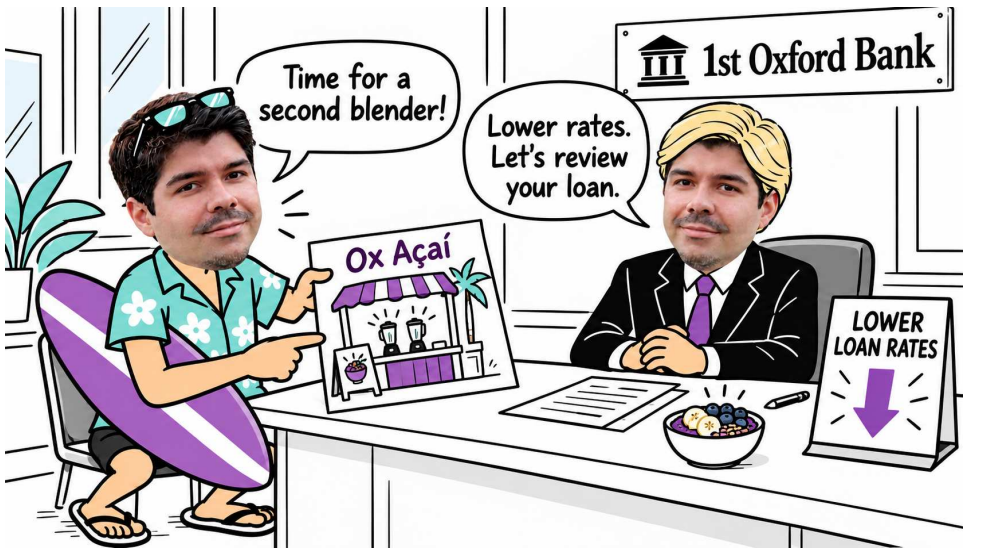
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Easier financing can support borrowing.

Bank reserves +$1,000 · Monetary base +$1,000

Bank securities −$1,000 · Public deposits: no immediate change.

Loan-rate effects are conditional; reserve changes alone do not guarantee them.

New York Fed, Monetary Policy Implementation

Open-Market Sales

The Fed sells a $1,000 Treasury security to a commercial bank.



Sale: reserves leave the bank.

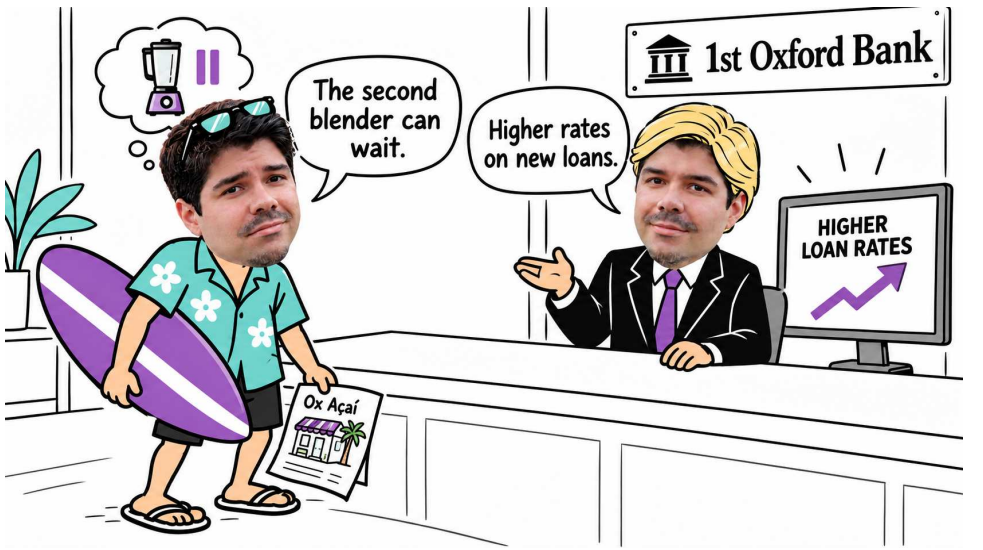
New York Fed, Monetary Policy Implementation

Open-Market Sales

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Costlier financing can postpone borrowing.

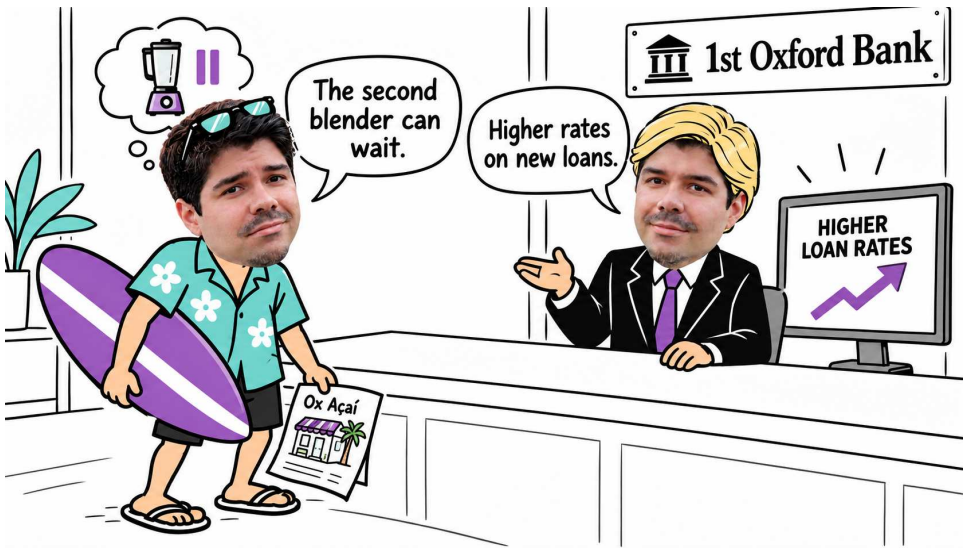
New York Fed, Monetary Policy Implementation

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Sale: reserves leave the bank.



Costlier financing can postpone borrowing.

Bank reserves −$1,000 · Monetary base −$1,000

Bank securities +$1,000 · Public deposits: no immediate change.

Loan-rate effects are conditional; reserve changes alone do not guarantee them.

New York Fed, Monetary Policy Implementation

The ECB and the Federal Reserve

ECB = European Central Bank · Eurosystem = ECB + euro-area national central banks.

	Federal Reserve	ECB / Eurosystem
Central board	7 Board governor seats	6 Executive Board members
Policy committee		
Operating network		
Main policy goals		

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

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Main policy goals	Maximum employment; stable prices	Price stability is primary

Both use policy rates, lending facilities, and asset operations.

The ECB’s deposit facility offers a comparable remuneration benchmark.

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

Reference Definitions and Examples

Money, income, wealth, and bank accounting.

Money, Income, and Net Wealth

Concept	Meaning	Example
Money	Readily spendable assets	$400 in currency and deposits
Income	Receipts over a period	$900 earned this month
Net wealth	Assets minus liabilities	$5,000 − $2,000 = $3,000

A Smoothie Company Balance Sheet

Assets	Liabilities and equity
Cash and inventory	Loans and other obligations
Equipment; a building if owned	Equity: the owners' residual claim

Assets = liabilities + equity.

Liquidity and Solvency

Issue	Question	Possible problem
Liquidity	Can the bank pay now?	Assets cannot all be sold quickly at little cost.
Solvency	Do assets cover liabilities?	Losses can exhaust equity.

Forced sales can turn liquidity pressure into losses.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

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Reserves and deposits each rise by $200.

Public currency falls by $200; currency + deposits is unchanged.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

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A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.
- The borrower gains a deposit and owes a debt; net wealth does not rise by $120.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.

Source: Federal Reserve Board, H.6 definitions.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.

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Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.
- **Monetary base**: public currency plus bank reserves.

Source: Federal Reserve Board, H.6 definitions.